

An Alternative to Chinese Remainder Theorem

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Find the smallest number , when divided by 3 , 5 and 7 leaves the remainder 1,2 and 3, respectively

Algorithm				Algorithm			
Divisors				3		5	7
Remainders				1		2	3
{3 − 1 = 2}				2		2	2
New Remainders				0		4	5
{3 × 2 = 6}				6		6	6
New Remainders				0		0	4
{3 × 5 × 3 = 45}				45		45	45
New Remainders				0		0	0
Required number = {lcm of 3,5,7} − {2 + 6 + 45}							
Required least number = {105 − 53} = 52							
{210 − 53} = 157 Will also satisfy the conditions.							
Verification:				Verification:			
Divisor	Number	Remainder		Divisor	Number	Remainder	
3	157	1.Exactly		3	52	1.Exactly	
Divisor	Number	Remainder		Divisor	Number	Remainder	
5	157	2.Exactly		5	52	2.Exactly	
Divisor	Number	Remainder		Divisor	Number	Remainder	
7	157	3.Exactly		7	52	3.Exactly	

Verification:

Divisor	Number	Remainder
5	366	1.Exactly
Divisor	Number	Remainder
7	366	2.Exactly
Divisor	Number	Remainder
11	366	3.Exactly

Find the smallest number , when divided by 5 , 7 and 11 leaves the remainder 1,2 and 3, respectively

Algorithm	Algorithm		
Divisors	5	7	11
Remainders	1	2	3
$\{5 - 1 = 4\}$	4	4	4
New Remainders	0	6	7
$\{5 \times 3 = 15\}$	15	15	15
New Remainders	0	0	0
$\{3 \times 5 \times 3 = 45\}$	0	0	0
New Remainders	0	0	0
<i>Required number = {lcm of 5,7,11} - {4 + 15}</i>			
<i>Required least number = {385 - 19} = 366</i>			

Verification:

Divisor	Number	Remainder
5	366	1.Exactly
Divisor	Number	Remainder
7	366	2.Exactly
Divisor	Number	Remainder
11	366	3.Exactly

Successive Correction Algorithm for Congruences

Let

$$x \equiv r_i \pmod{d_i}, \quad i = 1, 2, \dots, n,$$

where

$$d_1, d_2, \dots, d_n$$

are pairwise coprime positive integers and

$$0 \leq r_i < d_i.$$

Define

$$L = \text{lcm}(d_1, d_2, \dots, d_n).$$

Let

$$c_1 = d_1 - r_1.$$

For each $k \geq 2$, define

$$c_k = (d_1 d_2 \cdots d_{k-1}) x_k,$$

where x_k is chosen so that

$$c_1 + c_2 + \cdots + c_k \equiv d_k - r_k \pmod{d_k}.$$

Then

$$C = c_1 + c_2 + \cdots + c_n$$

satisfies

$$C \equiv d_i - r_i \pmod{d_i}, \quad i = 1, \dots, n,$$

and therefore

$$N = L - C$$

satisfies

$$N \equiv r_i \pmod{d_i}, \quad i = 1, \dots, n.$$

Hence

$$N = \text{lcm}(d_1, d_2, \dots, d_n) - \sum_{k=1}^n c_k$$

is a solution of the given system of congruences.

For $k = 1$,

$$c_1 = d_1 - r_1,$$

so

$$C_1 = c_1 \equiv d_1 - r_1 \pmod{d_1}.$$

Assume that after step $k - 1$,

$$C_{k-1} = c_1 + c_2 + \dots + c_{k-1}$$

satisfies

$$C_{k-1} \equiv d_i - r_i \pmod{d_i}, \quad (i < k).$$

Choose

$$c_k = (d_1 d_2 \cdots d_{k-1}) x_k$$

so that

$$C_k = C_{k-1} + c_k \equiv d_k - r_k \pmod{d_k}.$$

Since

$$c_k$$

is divisible by

$$d_1, d_2, \dots, d_{k-1},$$

we have

$$c_k \equiv 0 \pmod{d_i}, \quad (i < k).$$

Hence

$$C_k \equiv C_{k-1} \equiv d_i - r_i \pmod{d_i}, \quad (i < k).$$

Thus all previously established congruences remain unchanged, while the k -th congruence is satisfied. By induction,

$$C = C_n$$

satisfies

$$C \equiv d_i - r_i \pmod{d_i}, \quad i = 1, \dots, n.$$

Since

$$L = \text{lcm}(d_1, d_2, \dots, d_n)$$

is divisible by every d_i ,

$$L \equiv 0 \pmod{d_i}.$$

Therefore

$$N = L - C \equiv r_i \pmod{d_i}, \quad i = 1, \dots, n.$$

Hence N satisfies all the required congruences.

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively

Algorithm	Algorithm				
Divisors	d_1	d_2	d_3	d_4	d_5
Remainders	r_1	r_2	r_3	r_4	r_5
$(d_1 - r_1) = a_1$	a_1	a_1	a_1	a_1	a_1
New Remainder	0	a_2	a_3	a_4	a_5
$(d_1)(x_1) = a_2$	0	a_2	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	0	0	0	0	0
Multiply by suitable values to make all remainders zero					
Required least number = $\Pi_{i=1}^n \{d_i\} - \{\sum_{i=1}^n a_i\}$	m_1				
Multiplier 2	m_2				
New Values					
Multiplier 3	m_3				
New Values					
Required least number = $\text{LCM}\{d_1, d_2, d_3, d_4, d_5\} - \sum_{i=1}^n a_i$					
Required least number = $\Pi_{i=1}^n \{d_i\} - \{\sum_{i=1}^n a_i\}$	0				
New Values	a_1	a_2	a_3	a_4	a_5

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively

Algorithm	Algorithm				
Divisors	d_1	d_2	d_3	d_4	d_5
Remainders	r_1	r_2	r_3	r_4	r_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
Multiply suitable values to make most columns zero					
Multiplier 1	m_1				
New Values					
Multiplier 2	m_2				
New Values					
Multiplier 3	m_3				
New Values					
Required number = $\text{LCM}(d_1, d_2, d_3, d_4, d_5) - (a_1 + m_1 + m_2 + m_3)$					

An Alternative to the Chinese Remainder Theorem

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively.

Divisors	3	5	7	11	13
Remainders	2	3	2	1	10
Initial Correction	1	1	1	1	1
Residual Values	0	1	4	9	2
Correction	0	6	6	6	6
Residual Values	0	0	5	3	9
Correction	0	0	75	75	75
Residual Values	0	0	0	5	12
Correction	0	0	0	1050	1050
Residual Values	0	0	0	0	2
Correction	0	0	0	0	13860
Final Residuals	0	0	0	0	0

$$L = \text{LCM}(3, 5, 7, 11, 13) = 15015$$

$$C = 1 + 6 + 75 + 1050 + 13860 = 14992$$

$$N = L - C = 15015 - 14992 = 23$$

$N = 23$ **Verification:**

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively

Algorithm	Algorithm				
Divisors	d_1	d_2	d_3	d_4	d_5
Remainders	r_1	r_2	r_3	r_4	r_5
$(d_1 - r_1) = a_1$	a_1	a_1	a_1	a_1	a_1
New Remainder	0	a_2	a_3	a_4	a_5
$(d_1)(x_1) = a_2$	0	a_2	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	a_1	a_2	a_3	a_4	a_5
$d_i - r_i$	$d_1 - r_1$	$d_2 - r_2$	$d_3 - r_3$	$d_4 - r_4$	$d_5 - r_5$
New Values	0	0	0	0	0
Multiply by suitable values to make all remainders zero					
Required least number = $\Pi_{i=1}^n \{d_i\} - \{\sum_{i=1}^n a_i\}$	m_1				
Multiplier 2	m_2				
New Values					
Multiplier 3	m_3				
New Values					
Required least number = $\text{LCM}\{d_1, d_2, d_3, d_4, d_5\} - \sum_{i=1}^n a_i$					
Required least number = $\{\Pi_{i=1}^n d_i\} - \{\sum_{i=1}^n a_i\}$ When all new values become	0	0	0	0	0

An Alternative to the Chinese Remainder Theorem by Prajay S.Thul

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively.

Divisors	3	5	7	11	13
Remainders	2	3	2	1	10
Initial Correction	1	1	1	1	1
Residual Values	0	1	4	9	2
Correction	0	6	6	6	6
Residual Values	0	0	5	3	9
Correction	0	0	75	75	75
Residual Values	0	0	0	5	12
Correction	0	0	0	1050	1050
Residual Values	0	0	0	0	2
Correction	0	0	0	0	13860
Final Residuals	0	0	0	0	0

$$L = \text{LCM}(3, 5, 7, 11, 13) = 15015$$

$$C = 1 + 6 + 75 + 1050 + 13860 = 14992$$

$$N = L - C = 15015 - 14992 = 23$$

$N = 23$ **Verification:**

An Alternative to the Chinese Remainder Theorem

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively.

Divisors	d_1	d_2	d_3	d_4	d_5
Remainders	r_1	r_2	r_3	r_4	r_5
Initial Correction	$d_1 - r_1$				
Residual Values	0	b_2	b_3	b_4	b_5
Correction	0	$c_2 = d_1 x_2$			
Residual Values	0	0	b'_3	b'_4	b'_5
Correction	0	0	$c_3 = d_1 d_2 x_3$		
Residual Values	0	0	0	b''_4	b''_5
Correction	0	0	0	$c_4 = d_1 d_2 d_3 x_4$	
Residual Values	0	0	0	0	b'''_5
Correction	0	0	0	0	$c_5 = d_1 d_2 d_3 d_4 x_5$
Final Residuals	0	0	0	0	0

$$L = \text{LCM}(d_1, d_2, d_3, d_4, d_5)$$
$$C = (d_1 - r_1) + c_2 + c_3 + c_4 + c_5$$
$$N = L - C$$

Find the smallest number , when divided by 3 , 5 and 7 leaves the remainder 1,2 and 3, respectively

Algorithm				Algorithm			
Divisors				3		5	7
Remainders				1		2	3
$\{3 - 1 = 2\}$				2		2	2
New Remainders				0		4	5
$\{3 \times 2 = 6\}$				6		6	6
New Remainders				0		0	4
$\{3 \times 5 \times 3 = 45\}$				45		45	45
New Remainders				0		0	0
$Required\ number = \{lcm\ of\ 3,5,7\} - \{2 + 6 + 45\}$							
$Required\ least\ number = \{105 - 53\} = 52$							
$\{210 - 53\} = 157$ Will also satisfy the conditions.							
Verification:				Verification:			
Divisor	Number	Remainder		Divisor	Number	Remainder	
3	157	1.Exactly		3	52	1.Exactly	
Divisor	Number	Remainder		Divisor	Number	Remainder	
5	157	2.Exactly		5	52	2.Exactly	
Divisor	Number	Remainder		Divisor	Number	Remainder	
7	157	3.Exactly		7	52	3.Exactly	

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively

Table 1: Algorithm

Step	d_1	d_2	d_3	d_4	d_5
Divisors	d_1	d_2	d_3	d_4	d_5
Remainders	r_1	r_2	r_3	r_4	r_5
$c_1 = d_1 - r_1$	c_1	c_1	c_1	c_1	c_1
Partial Sum					
remainders					
$c_2 = (d_1) \cdot x_2$	c_2				
Partial Sum					
remainders					
$c_3 = (d_1 d_2) \cdot x_3$	c_3				
Partial Sum					
remainders					
$c_4 = (d_1 d_2 d_3) \cdot x_4$	c_4				
Partial Sum					
remainders 0	0		0	0	0
$c_5 = (d_1 d_2 d_3 d_4) \cdot x_5$	c_5				
Total C	$C = \sum c_k$				
$\mathbf{L} = \text{lcm}(d_1, d_2, \dots, d_5)$					
Required number $N = \mathbf{L} - \mathbf{C}$					

Table 2: Verification

Divisor	N	Remainder (must equal r_i)
d_1	N	r_1
d_2	N	r_2
d_3	N	r_3
d_4	N	r_4
d_5	N	r_5

Find the smallest number , when divided by 3 , 5 and 7 leaves the remainder 1,2 and 3, respectively

Table 3: Algorithm

Step	3	5	7
Divisors	3	5	7
Remainders	1	2	3
$c_1 = 3 - 1$	2	2	2
Partial Sum	2	2	2
$c_2 = 3 \times 2 = 6$	6		
Partial Sum	8	8	8
$c_3 = 15 \times 3 = 45$	45		
Total C	53		
$L = 105$			
Required number= $105-53=52$			

Table 4: Verification:

Divisor	Number	Remainder
3	52	1
5	52	2
7	52	3

An Alternative to the Chinese Remainder Theorem

Find the smallest number which when divided by d_1, d_2, d_3, d_4, d_5 leaves remainders r_1, r_2, r_3, r_4, r_5 respectively.

Divisors	d_1	d_2	d_3	d_4	d_5
Remainders	r_1	r_2	r_3	r_4	r_5
Initial Correction	$d_1 - r_1$				
Residual Values	0	b_2	b_3	b_4	b_5
Correction	0	$c_2 = d_1 x_2$			
Residual Values	0	0	b'_3	b'_4	b'_5
Correction	0	0	$c_3 = d_1 d_2 x_3$		
Residual Values	0	0	0	b''_4	b''_5
Correction	0	0	0	$c_4 = d_1 d_2 d_3 x_4$	
Residual Values	0	0	0	0	b'''_5
Correction	0	0	0	0	$c_5 = d_1 d_2 d_3 d_4 x_5$
Final Residuals	0	0	0	0	0

$$L = \text{LCM}(d_1, d_2, d_3, d_4, d_5)$$

$$C = (d_1 - r_1) + c_2 + c_3 + c_4 + c_5$$

$$N = L - C$$